
To Bind or Not to Bind: A Layer-Wise Dissection of Binding Information in ESM

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Manuscript — 2025

Abstract

Large protein language models (PLMs) are increasingly used to study and design proteins, yet it remains unclear where and how they encode biologically relevant information. We investigate this question through the case study of TCR CDR3b–peptide binding, probing the Evolutionary Scale Model (ESM) with a combination of difference-of-means analyses, classifier training on intermediate embeddings, and causal ablation of attention heads. Our findings suggest that intermediate representations may be more informative than final embeddings for binding prediction, while also highlighting the broader challenge of defining meaningful interpretability criteria for biological models and motivating future work on methods that can bridge internal signals to biologically legible concepts.

1 Introduction

Protein language models (PLMs) have transformed protein science by learning rich sequence representations that enable accurate prediction of structure and function [Rives et al., 2021, Elnaggar et al., 2021]. Early efforts to interpret these models often sought to extract the underlying biological patterns in the model "knowledge". For instance, attention analyses suggested that PLMs capture residue–residue contacts and other structural relationships [Vig et al., 2020]. More recently, sparse autoencoders (SAEs) have emerged as a promising tool for disentangling polysemantic activations, enabling the discovery of interpretable latent features.

Applied to PLMs, SAEs have been shown to recover features that align with biological concepts such as binding sites, motifs, and domains [Simon and Zou, 2024]. Intervening on these latents can even steer model behavior toward desired motifs and structural properties [Villegas Garcia and Ansuini, 2025, Parsan et al., 2025].

Yet, a critical challenge remains. Unlike in natural language or vision, where outputs are human-legible, the "semantics" of proteins (structural folds, binding affinities, thermostability, etc.) are not directly visible in sequence space. One cannot simply read an amino acid sequence and infer its function in the way one can read a sentence and recognize tone [Ko et al., 2008]. This fundamental gap between interpretable ground truth and model outputs makes mechanistic interpretability in biology uniquely difficult.

In this study, we take a step toward closing this gap by probing the Evolutionary Scale Model (ESM) through a proxy classifier. Specifically, we focus on the TCR CDR3b–peptide binding task, as studied by [Yadav et al., 2024] and illustrated in Figure 1. We examine how embeddings produced by ESM

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Using Evolutionary Scaling Model (ESM) embeddings to predict TCR CDR3b-peptide binding

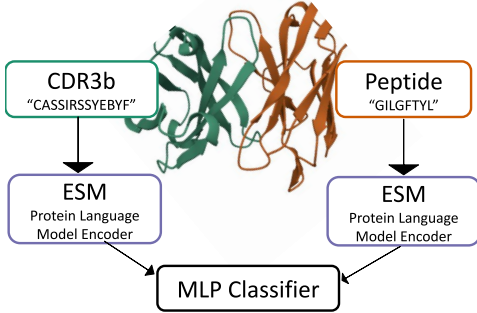


Figure 1: Schematic of the TCR CDR3b-peptide binding prediction task. CDR3b and peptide sequences are separately encoded with ESM (mean-pooled over tokens), and the resulting embeddings are concatenated and passed into an MLP classifier to predict binding versus non-binding interactions.

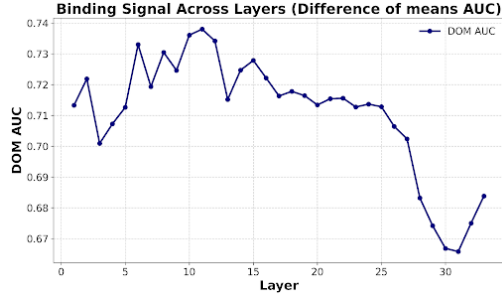


Figure 2: Binding signal across ESM layers using a difference-of-means (DoM) probe. For each layer, we defined a “binding direction” as the difference between mean binder and non-binder embeddings, projected sequences onto this direction, and evaluated separation by AUC. Signal peaks in early-to-mid layers (AUC = 0.74) and declines in deeper layers, indicating that binding information is most accessible in intermediate representations.

relate to the ability to distinguish binders from non-binders. This proxy task provides a testbed to explore a central question: how is binding information encoded in PLMs? Are particular layers, MLPs, or heads especially relevant? And can such analyses yield models that are both more accurate and more interpretable, and thus more useful for biological researchers?

2 TCR CDR3b-peptide Binding Task Setup

We adopt the experimental setup from [Yadav et al., 2024] to ensure comparability with prior work on TCR-pMHC binding. The NetTCR2.0 dataset serves as our primary training corpus, consisting of positive CDR3b-peptide binding pairs curated from VDJdb and McPAS-TCR, with negatives generated via mismatching as described in the original study. Further details on training, the ESM protein language model, and the multi-layer perceptron classifier are provided in Appendix 7.1.

2.1 Evaluation Protocol

Performance is evaluated on the held-out MIRA test set, which contains over 1,000 CDR3b-peptide pairs, consistent with the NetTCR2.0 and TCR-ESM benchmarks. Following [Yadav et al., 2024], we report the Matthews correlation coefficient (MCC) as our primary evaluation metric, since it provides a balanced assessment of performance on imbalanced datasets [Chicco and Jurman, 2020, Chicco et al., 2021]. Code and experiment scripts are available at:

<https://github.com/kevinlu4588/ESMProbing>

3 Exploring ESM Layers

To begin, we probed the ESM model to examine how binding signal is distributed across layers. As an initial training-free method, we applied a simple *difference of means* (DoM) analysis [Marks and Tegmark, 2024]. For each layer, we computed the mean embedding of binders and non-binders, defined their difference as a candidate “binding direction,” and projected sequence embeddings onto this vector. As shown in Fig. 2, this probe reveals the strongest signal in early and middle layers, with some degradation toward the final layers.

In addition to the DoM probe, we also evaluated an L_2 -norm variant and conducted an activation patching experiment (Appendix 7.2). These analyses suggest that model importance is concentrated in early layers (for DoM) and late layers (for patching), highlighting distinct phases of computation.

Nonetheless, the base classifier from prior work was trained on the final embeddings. The DoM results therefore raise a natural question: *what happens if we train a classifier directly on intermediate embeddings from non-final layers of ESM?*

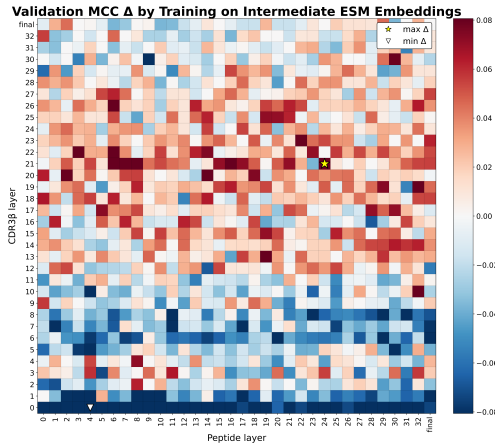


Figure 3: For each cell, an MLP classifier was trained on CDR3b embeddings from the row layer and peptide embeddings from the column layer. Colors indicate Δ validation MCC compared to the baseline classifier trained on final-layer embeddings. Overall, intermediate layers often yield improved generalization compared to the final-layer representations.

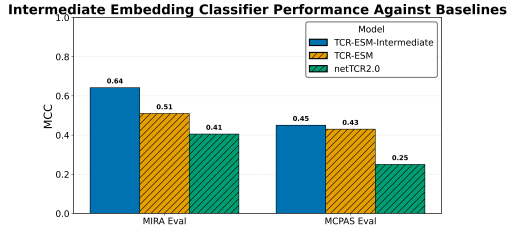


Figure 4: Comparison of classifier performance across datasets. On the MIRA evaluation set, the intermediate-embedding classifier (TCR-ESM-Intermediate) achieves substantially higher MCC (0.64) compared to the base TCR-ESM (0.51) and netTCR2.0 (0.41). On the external MCPAS dataset, improvements are more modest (0.45 vs. 0.43 and 0.25), suggesting stronger gains on in-distribution data but limited transfer to a distinct evaluation set.

4 Linear Probing Enhances Binding Prediction Signal

For the training process, we constructed datasets of embeddings from intermediate layers of ESM for both the CDR3b sequences and their associated peptides. We then trained multi-layer perceptron (MLP) classifiers on different combinations of embeddings—selecting one layer for the CDR3b and another for the peptide—and evaluated their performance.

As shown in Fig. 3, classifiers trained on intermediate embeddings often achieve improved training loss and higher MCC scores on the external test set compared to classifiers trained on final-layer embeddings. Interestingly, rather than observing a simple diagonal pattern of optimal performance (where both inputs come from the same layer), we find a distinct “band” of better-performing classifiers, largely determined by the choice of CDR3b layer. This observation aligns with biological intuition: in TCR–pMHC binding, most of the sequence variability and functional specificity arises from the CDR3 region [Dash et al., 2017, Glanville et al., 2017].

We compare the model with the best validation MCC (trained on embeddings from layer 21 for CDR3b sequences and layer 24 for peptide sequences), to the base model and other baselines (Fig. 4). On the MIRA dataset, we observe a significant improvement in test MCC, while the MCPAS dataset shows only a modest gain. Although the signal on MCPAS is weaker, these results suggest that training on intermediate embeddings (rather than relying solely on later layers) can still yield meaningful benefits. Supporting this view, we also find similar improvements when applying the same approach to ESM-2, as detailed in Appendix 7.3.

5 Causal Probing of Late-Layer Attention Heads

To probe the later layers in more detail, we first collect sequences from the MIRA test set where the new intermediate-layer classifier makes correct predictions but the base classifier fails. As shown in Fig. 5, four heads in the final layers exhibit especially strong activations. We then adopt a causal approach by ablating these heads and examining the effect on downstream performance (Fig. 6).

Interestingly, ablating these “incorrect” heads in the ESM model and then passing the modified embeddings into the base classifier improves MCC on the test set without any additional training.

This improvement is consistently larger than the best performance obtained by ablating one to four random heads. When applying the same technique to the MCPAS dataset, however, the gains are weaker, as analyzed in 8.1. We also include additional representation-similarity analyses in Appendix 8, as well as experiments ablating units in the MLP layers, which we found uniformly degraded performance across the board.

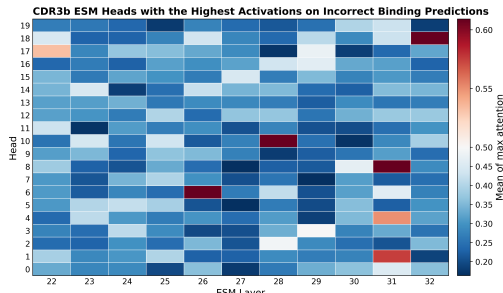


Figure 5: Mean of maximum attention values across heads (y-axis) and layers (x-axis) for CDR3b sequences, computed on MIRA test cases where the new intermediate-layer classifier predicts correctly but the base classifier fails. Several heads in later layers show consistently elevated activations, suggesting they may be responsible for the degradation of useful representations.

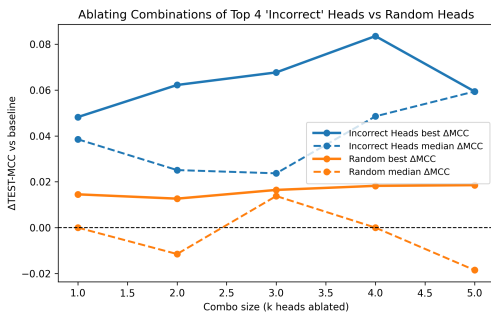


Figure 6: The four most "incorrect" heads were ablated in all combinations (blue), showing consistent improvements in MCC over the baseline without retraining. In contrast, ablating random sets of heads (orange) did not yield similar gains, highlighting a causal role of these heads in harmful mispredictions.

6 Discussion and Limitations

The goal of this study was to investigate where in ESM binding-relevant information is encoded. While we relied on a classifier as a proxy interpreter, this approach nonetheless revealed meaningful internal signals. Difference-of-means probing and classifier validation showed that intermediate representations capture binding information more clearly than final-layer embeddings. Finally, causal mediation analysis of attention heads in the later layers suggested that certain heads are directly associated with degraded predictive performance. At the same time, these signals varied in strength and consistency across datasets and probing methods (Fig.4, Appendix7.2, Appendix 8.1), highlighting the difficulty of drawing stable conclusions from any single interpretability technique.

More importantly, some questions remain unresolved. For example, what is the true role of the heads highlighted in Fig. 5? More broadly, how should we understand and describe the phenomena, features, and neurons that we uncover in these models? In our case, we attempted to use a classifier as an "interpreter," translating ESM embeddings into binding logits that humans can understand. In parallel, work like InterPLM has explored sparse autoencoders combined with LLM-based "autointerp" methods, where features and examples are passed to a model like Claude Sonnet to generate human-readable labels.

Yet both approaches face limitations: classifiers are only as good as their training data, and LLM-based annotation pipelines require richly annotated datasets, which are often lacking in biological contexts. This raises a broader question: must we always convert learned features into immediate human semantic outputs? Perhaps instead we can lean into the "language" of the models themselves. Biological models already capture patterns beyond our current understanding of evolution, sequence, and structure. By comparing the internal representations of models across modalities, we may discover shared abstractions that function as a kind of common vocabulary for biology.

[Dravid et al., 2023]’s work on "Rosetta neurons" in vision models illustrates this possibility: they found neurons shared across different architectures (generative, discriminative, etc.) that provided a unifying lens on model computations. Similarly, in protein modeling, one could imagine linking ESM attention heads to structural regions revealed by AlphaFold, or discovering features conserved across protein sequence and structure models. Such cross-modality analysis could illuminate "fundamental concepts" in biology and provide a foundation for mechanistic interpretability.

This study was an exploratory step toward understanding the internals of protein language models, while also highlighting the difficulties inherent to interpreting them. Neuroscience has long faced a similar challenge: neural activity is not directly human-readable, and meaning must be inferred through probing, ablation, and cross-modal comparison[Devereux et al., 2018]. Linking sequence, structure, and function across modalities could bring us closer to a mechanistic understanding of learned biological representations.

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7 Appendix A

7.1 Model setup

NetTCR2.0 dataset

The NetTCR2.0 datasets consist of over 30,000 CDR3b and peptide pairs. Approximately 20 percent of the pairs are "positive" binders, and the other 80 percent are synthetically generated negative binders. This is common practice within binding datasets due to the narrow specificity of binding interactions between proteins.

Embedding Extraction

We used ESM-1v, a large protein language model trained on UniRef90 sequences, to obtain fixed-length representations of CDR3b and peptide sequences. Following the TCR-ESM protocol, we extracted the penultimate layer embeddings (dimension 1280 per residue) using the fair-esm Python library. To obtain sequence-level vectors, we applied global average pooling across residues, yielding one 1280-dimensional embedding per sequence.

Model Architecture

For binding prediction, we trained a multi-layer perceptron (MLP) classifier. The TCR and peptide embeddings were concatenated and fed into the network. Hidden layers used GELU activations and dropout regularization. We optimized hyperparameters (learning rate, hidden layer size, dropout rate) through grid search on the NetTCR training split. Training was performed using binary cross-entropy loss with Adam optimization and a learning-rate scheduler that decayed on validation plateaus.

7.2 Additional Probing Results

In addition to using the difference-of-means vector to calculate AUC-ROC scores, we also examined the raw L2 norms of the difference-of-means vectors and their z -scored counterparts across layers. Both figures in Fig. 7 exhibit the same overall pattern: the magnitude of the difference-of-means vector increases steadily with depth, followed by a sharp drop at the final layer. We interpret this final drop as likely being an artifact of the representation rather than a genuine loss of signal.

We also applied an activation patching setup to further probe how binding information is distributed across layers Fig 8. Specifically, we selected positive and negative binding pairs and patched the activations from the positive examples into the negative ones, token by token and layer by layer. We then measured the average change in model logits as a function of layer depth. Interestingly, this analysis revealed a different notion of "layer importance": the early layers produced larger logit changes under activation patching, in contrast to the steadily increasing signal observed with the difference-of-means analysis.

Together, these experiments highlight complementary perspectives on where binding information is encoded within ESM. Nonetheless, we found that the DoM AUC curve provided the most robust signal and therefore motivated our deeper investigation of the central layers.

7.3 ESM2 Intermediate Classifier Training Results

To test whether the patterns we observed in ESM also hold in newer architectures, we repeated our probing experiments on ESM2, a larger and more recent version of ESM. The heatmap (Fig. 9) shows

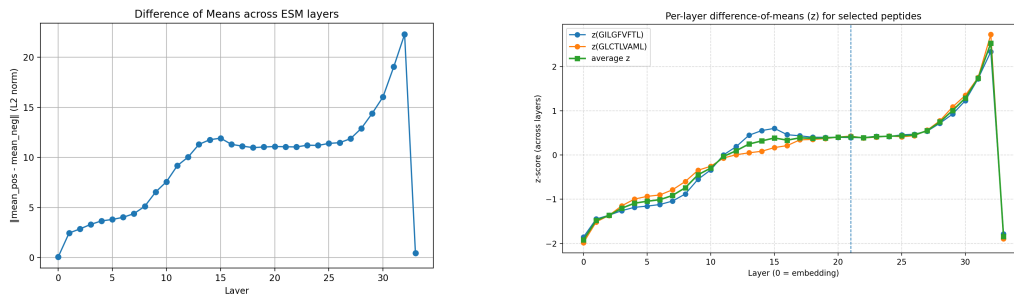


Figure 7: Per-layer difference-of-means analyses. (7) L2 norm of the difference between binder and non-binder mean embeddings across the entire dataset. (8) z -scored curves for two specific peptides (GILGFVFTL and GLCTLVAML), showing that peptide-specific trends largely follow the global trajectory, with signal strongest in later layers and a sharp drop at the final representation.

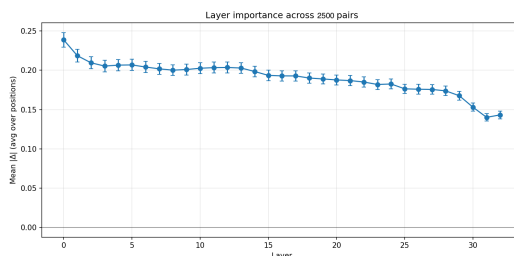


Figure 8: Activation patching analysis of layer importance. We patched activations from positive binding examples into negative ones and measured the average change in model logits across 2,500 sequence pairs. The largest changes occur in early layers, suggesting that binding-relevant information is strongly encoded at shallow depths, in contrast to the steadily increasing separation observed with difference-of-means probing.

that intermediate embeddings again outperform final-layer representations when used as inputs to a downstream classifier. While performance varies across different CDR3–peptide layer combinations, the strongest gains occur when CDR3 embeddings are taken from early-to-mid layers and paired with peptide embeddings from later layers, suggesting that complementary information is distributed across the network depth.

We then trained classifiers on these intermediate embeddings and evaluated them against baselines (Fig. 10). Across both the MIRA and MCPAS datasets, the best-performing intermediate models (TCR-ESM2-Inter) achieve higher MCC than both final-layer ESM2 embeddings and prior state-of-the-art methods such as netTCR2.0. This confirms that the phenomenon of intermediate representations capturing binding-relevant features is robust to model scale and architecture.

8 Appendix B

8.1 MCPAS Head Ablation Study

To assess the robustness of our causal probing, we repeated the head ablation experiments on the MCPAS dataset. As shown in Fig. 11, ablating the top four “incorrect” heads again yielded modest gains in test MCC relative to baseline, but the signal was noticeably weaker and more variable than on MIRA. While certain combinations of heads improved performance beyond random ablations, the overall trend was less consistent, suggesting that the role of these heads may be more dataset-dependent. This highlights both the potential and the limitations of using head ablations as causal probes across different evaluation settings.

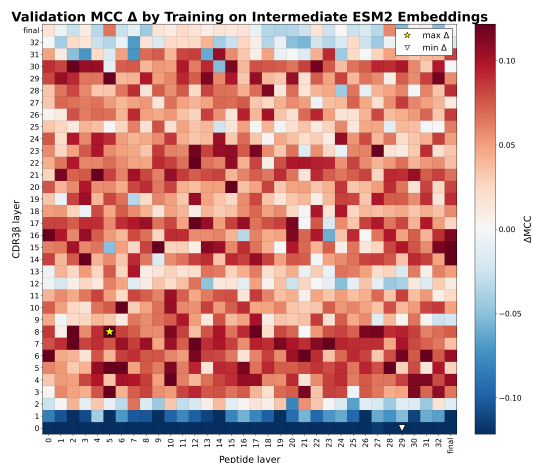


Figure 9: Validation MCC improvement (Δ) from training classifiers on intermediate ESM2 embeddings. Each cell shows the change in MCC relative to using final-layer embeddings, with rows corresponding to CDR3 layers and columns to peptide layers. Warm colors indicate performance gains, while cool colors indicate losses. The strongest gains occur when pairing early-to-mid CDR3 layers with later peptide layers.

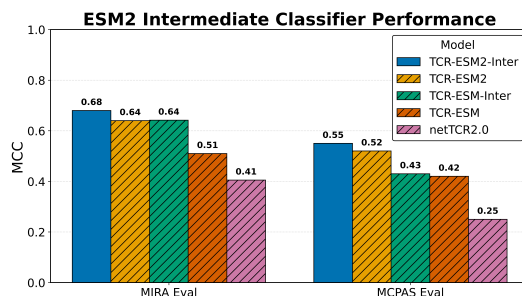


Figure 10: Comparison of classifier performance using intermediate ESM2 embeddings versus baselines. Bars show MCC on the MIRA and MCPAS evaluation sets. The best intermediate model (TCR-ESM2-Inter) outperforms both final-layer ESM2 embeddings and prior state-of-the-art methods (TCR-ESM, netTCR2.0), confirming the value of intermediate representations for binding prediction.

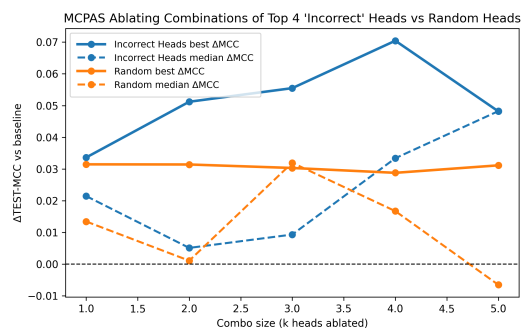


Figure 11: Effect of ablating the four most activated late-layer heads on the MCPAS dataset. Gains are weaker and more variable than in MIRA, suggesting dataset-dependent roles for these heads.

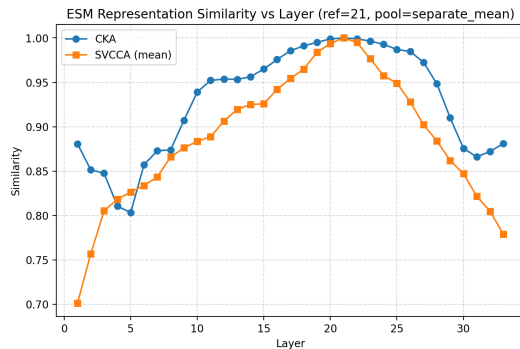


Figure 12: Layer-wise similarity of ESM representations relative to layer 21, measured using CKA and SVCCA. Both metrics reveal a strong band of similarity across mid-to-late layers, with divergence at the earliest and deepest layers. This suggests that intermediate layers form a stable representational core, while early and final layers encode more specialized information.

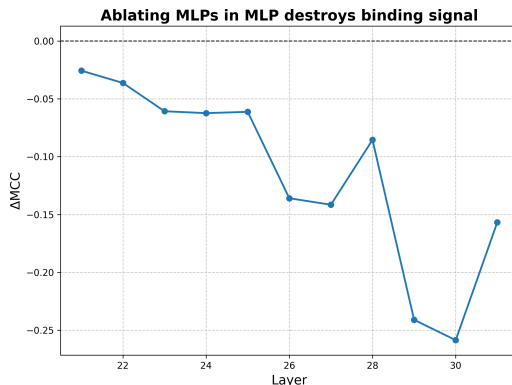


Figure 13: Ablating MLPs in ESM disrupts binding signal. Each point shows the change in Matthews correlation coefficient (MCC) after ablating the MLP at the indicated layer. Negative values indicate loss of binding information, with the largest effects occurring in later layers.

8.2 Embedding representation analysis

To better understand how information is distributed across the model, we examined the similarity of embeddings at different layers relative to a reference intermediate layer (layer 21). Using both CKA and SVCCA, we found that mid-to-late layers exhibit a strong band of representational similarity, whereas early and final layers diverge more substantially. This pattern suggests that intermediate layers form a stable core representation, while the earliest layers specialize in low-level features and the final layers reshape representations for task-specific outputs. Together, these results highlight the importance of intermediate embeddings as a robust source of biologically relevant signal.

8.3 MLP Ablation

To evaluate the contribution of the feed-forward sublayers, we systematically ablated the MLP readouts at each layer of ESM and measured the impact on classification performance. Figure 13 shows the change in Matthews correlation coefficient (MCC) relative to the intact model. Across layers, ablation consistently reduced performance, with the strongest degradation observed in the later layers. These results suggest that binding-relevant information is not only carried forward through the attention pathway but is also consolidated within the MLP blocks, and that disrupting these components substantially erodes the model’s ability to distinguish binders from non-binders.